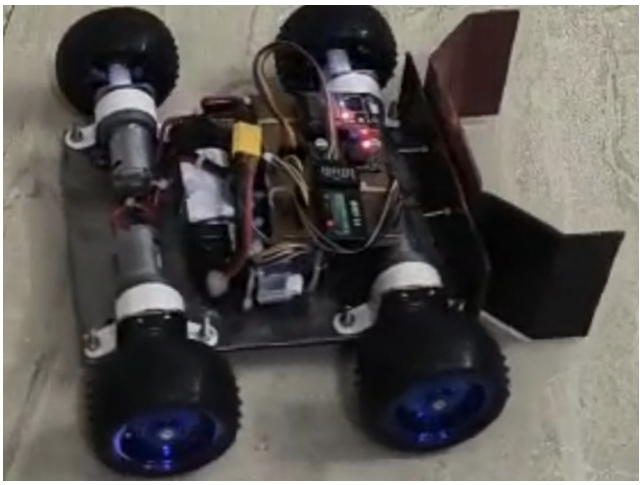
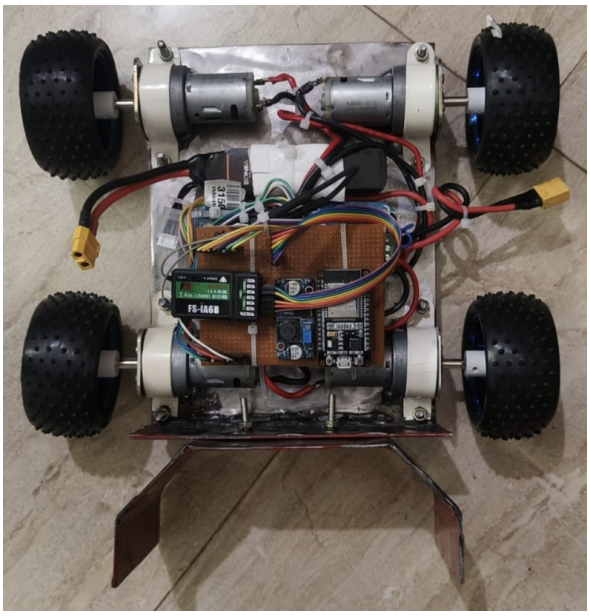
TARZAN

Multi-Purpose Robo Edge Bot

Team Sparks · Robo Edge · Robo Soccer / Robo Race / Obstacle Avoidance





Side / isometric view (left) · Top / electronics view (right)

1. MECHANICAL DESIGN

Dimensions (with mount)	30 × 30 × 5 cm
Dimensions (without mount)	23 × 30 × 5 cm
Weight	3 kg
Chassis Material	MS Steel
Chassis Type	Rigid
Wheel Configuration	4WD – 4 wheels
Wheel Diameter	85 mm
Ground Clearance	2 cm
Mount Shape	Half-hexagon, attached with nuts & bolts
Mount Angles	1st curve: 40° · 2nd curve: 45°
Mount Material	MS Steel
Mount Attachment	Fixed but removable

2. ELECTRONICS

Component	Specification	Qty	Purpose
ESP32 Dev Kit	3.3 V	1	Main control unit
BTS7960 Motor Driver	High-current driver	2	Motor control (one per side)

Johnson Geared Motor	1000 RPM	4	Drive wheels
LiPo Battery	11.1 V · 2200 mAh · 35C · 3S	1	Power supply
Buck Converter	—	1	Voltage regulation

3. CONTROL & COMMUNICATION

Transmitter	FlySky FS-i6 RC (2.4 GHz)
Receiver	FS-iA6B
Interface	Hardware RC – no smartphone app required
Control Range	Up to 1 km
Latency	No perceptible delay
Steering Method	Differential steering

4. PERFORMANCE

Parameter	Value
Top Speed	9 – 11 km/h
Current Load	3 kg
Max Load Capacity	6 kg (before motor stall)
Battery Runtime	~10 minutes

5. DESIGN RATIONALE & TRADE-OFFS

Motor Selection	High-RPM Johnson motors chosen to maximise speed for Robo Race and Obstacle Avoidance. Bot weight kept low to prevent stall, trading some torque for agility.
Wheel Choice	85 mm diameter knobby wheels deliver strong traction across competition surfaces.
Ground Clearance	2 cm clearance provides stable, controlled handling while remaining manageable over obstacles.
Motor Driver (BTS7960)	Handles heavy loads with lower heat generation compared to alternatives.
Front Mount	Half-hexagon MS steel scoop improves ball-pushing in soccer; minor trade-off in turning radius.

6. CHALLENGES & SOLUTIONS

Challenge	Resolution
Chassis fabrication — manually cutting and bending MS steel to precise angles (40° / 45°) while ensuring joints withstand collision impact.	Careful manual fabrication; reinforced bolt connections used throughout for structural integrity.
Wheel wobbling caused by 3D-printed hex couplers (metal equivalents unavailable).	Hot glue reinforcement combined with additional screws eliminated coupler play.
Power supply stability.	No issues encountered during testing.
Motor driver / motor reliability.	No issues encountered during testing.

7. FUTURE IMPROVEMENTS

Larger Wheels	Bigger diameter for improved grip and greater ground clearance — especially beneficial for Obstacle Avoidance.
Higher-Capacity Battery	Upgrade to larger capacity LiPo with higher C-rating for extended run time.
AI – Obstacle Avoidance	Integrate sensor-based or computer-vision AI for autonomous obstacle detection and avoidance.
AI – Line Following	Semi-autonomous line-following using computer vision.
AI – Ball Tracking	Computer-vision ball-tracking system to enhance Robo Soccer performance.